

Flat Modules and Tor

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Math 121

July 26, 2026

Much of this document is adapted from David Eisenbud's *Commutative Algebra: With a View Toward Algebraic Geometry* [Eis95]. Throughout this course, we have considered commutative rings with unity R , and modules over these rings. Out of these R -modules, we have also seen a variety of useful subcases: finitely-generated, free, cyclic, etc. However, one of these stands out among the rest. Indeed, *flat modules* are those making the functor $- \otimes_R M$ exact, and they allow for a rich theory that interacts with homological methods very cleanly.

Definition 1. We say that an R -module M is **flat** if the functor $- \otimes_R M$ is exact.

Remark 1. Note that since $M \otimes_R N$ is naturally isomorphic to $N \otimes_R M$, it does not matter whether we consider $- \otimes_R M$ or $M \otimes_R -$.

In particular, since we already know that tensor is right-exact, the flatness of M is equivalent to saying that tensoring by M preserves injections.

In practice, many of the nice modules we encounter such as R^n and $U^{-1}R$ are flat. We want to develop a way to detect whether a module is flat or not by studying various properties of the module. First, we have the following proposition:

Proposition 1. *For an R -module M , we have the following chain of implications:*

$$\text{free} \implies \text{projective} \implies \text{flat}.$$

Proof. We already saw in class that free modules are projective. Hence, it remains to prove that projective modules are flat. We first show that free implies flat then use this to prove projective implies flat.

Suppose that F is a free R -module. Then we can write

$$F \cong \bigoplus_{s \in S} R,$$

so we see that for any R -module M , we get

$$M \otimes_R F \cong M \otimes_R \left(\bigoplus_{s \in S} R \right) \cong \bigoplus_{s \in S} (M \otimes_R R) \cong \bigoplus_{s \in S} M.$$

Now, consider an injective map $f : A \hookrightarrow B$. The induced map $f \otimes \text{id}_F : A \otimes_R F \rightarrow B \otimes_R F$ under the identification above is given by the map $f' : \bigoplus_{s \in S} A \rightarrow \bigoplus_{s \in S} B$ where

$$f'(a_s)_{s \in S} = (f(a_s))_{s \in S}.$$

This map is injective since the map is injective componentwise, and hence, the induced map is injective for all injections, and F is flat.

Now let P be a projective R -module. Since P is projective, there exists an R -module Q such that

$$P \oplus Q \cong F$$

for some free R -module F . But we just saw that free modules are flat, so $- \otimes_R F$ is exact. Then,

$$- \otimes_R F \cong - \otimes_R (P \oplus Q) \cong (- \otimes_R P) \oplus (- \otimes_R Q),$$

and hence $- \otimes_R P$ is a direct summand of an exact functor, and must be exact. Hence, P is flat. $\square$

Example 1. The $\mathbb{Z}$ -module $\mathbb{Z}/2\mathbb{Z}$ is not flat. To see this, notice that

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

is exact, but after tensoring with $\mathbb{Z}/2\mathbb{Z}$, the map induced by $\cdot 2$

$$\mathbb{Z}/2\mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z}/2\mathbb{Z}$$

becomes the zero map, which is clearly not injective.

Now, we develop a collection of functors, called the Tor functors, which shall help us detect flatness. The Tor functors are examples of *derived functors*, which essentially measure the failure of a functor to be exact. First, we need a preliminary definition.

Definition 2. A **free resolution** of an R -module N is a chain complex of free modules $F_\bullet$

$$\cdots \longrightarrow F_{i+1} \xrightarrow{\varphi_{i+1}} F_i \xrightarrow{\varphi_i} F_{i-1} \xrightarrow{\varphi_{i-1}} \cdots \xrightarrow{\varphi_2} F_1 \xrightarrow{\varphi_1} F_0 \xrightarrow{\varphi_0} 0$$

together with an R -module homomorphism $\varepsilon : F_0 \rightarrow N$ making the following sequence exact:

$$\cdots \xrightarrow{\varphi_3} F_2 \xrightarrow{\varphi_2} F_1 \xrightarrow{\varphi_1} F_0 \xrightarrow{\varepsilon} N \longrightarrow 0.$$

Definition 3. Let M and N be R -modules. Consider a free resolution

$$\cdots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow 0$$

of N . Apply the functor $M \otimes_R -$ to obtain

$$\cdots \longrightarrow M \otimes_R F_2 \longrightarrow M \otimes_R F_1 \longrightarrow M \otimes_R F_0 \longrightarrow 0.$$

We define

$$\text{Tor}_i^R(M, N) := H_i(M \otimes_R F_\bullet) = \frac{\ker(M \otimes_R F_i \rightarrow M \otimes_R F_{i-1})}{\text{im}(M \otimes_R F_{i+1} \rightarrow M \otimes_R F_i)}.$$

Example 2. Let us first compute $\text{Tor}_0^R(M, N)$ for two R -modules M and N . Indeed,

$$\text{Tor}_0^R(M, N) = \frac{M \otimes_R F_0}{\text{im}(M \otimes_R F_1 \rightarrow M \otimes_R F_0)} = \text{coker}(M \otimes_R F_1 \rightarrow M \otimes_R F_0).$$

But recall that from the original resolution

$$\cdots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow N \longrightarrow 0$$

is exact, so that

$$\text{coker}(F_1 \rightarrow F_0) \cong N.$$

Now, $M \otimes_R -$ is right exact, so it preserves cokernels, and hence

$$\text{Tor}_0^R(M, N) = \text{coker}(M \otimes_R F_1 \rightarrow M \otimes_R F_0) = M \otimes_R \text{coker}(F_1 \rightarrow F_0) \cong M \otimes_R N.$$

Remark 2. Note that there is some ambiguity in our definition of Tor. First, we have not yet proved that every module has a free resolution for us to calculate Tor. Second, we also have not justified why Tor remains unchanged after considering different free resolutions of N , so that we can take any convenient resolution of N to compute Tor. Furthermore, we also have not yet proved that the Tor modules remain unchanged if we instead consider a free resolution of M instead of N . The fact that one can resolve either input of the Tor functor is called the “balancedness” of Tor, and we shall take this fact and the existence/independence of free resolutions on faith in this document.

Also, note that $\text{Tor}_i^R(M, N) \cong \text{Tor}_i^R(N, M)$ by the commutativity (up to natural isomorphism) of $- \otimes_R -$.

When dealing with homology, we often like to generate long exact sequences from short exact sequences that allow us to calculate the modules from information that we already know. Tor has an analogous computational tool, known as the **long exact sequence of Tor**. Indeed, for any short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ of R -modules, and any R -module M , we get the following long exact sequence from the long exact sequence for homology applied to the tensorized free resolution.

$$\begin{array}{ccccccc} \cdots & \longrightarrow & \text{Tor}_i^R(M, A) & \longrightarrow & \text{Tor}_i^R(M, B) & \longrightarrow & \text{Tor}_i^R(M, C) \\ & & & & \cdots & & \\ & & & & \text{Tor}_1^R(M, A) & \longrightarrow & \text{Tor}_1^R(M, B) & \longrightarrow & \text{Tor}_1^R(M, C) \\ & & & & & & & & \\ & & & & & & & & M \otimes_R A & \longrightarrow & M \otimes_R B & \longrightarrow & M \otimes_R C & \longrightarrow & 0. \end{array}$$

We are now ready to see how Tor is relevant to deciding the flatness of a module.

Proposition 2. *Let R be a ring, and let M be an R -module. M is flat iff $\text{Tor}_1^R(M, N) = 0$ for all R -modules N .*

Proof. First suppose that M is flat. Let N be any R -module, and choose a free resolution of N :

$$\cdots \longrightarrow F_2 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow N \longrightarrow 0.$$

Since M is flat, tensoring with M preserves exactness, so

$$\cdots \longrightarrow M \otimes_R F_2 \longrightarrow M \otimes_R F_1 \longrightarrow M \otimes_R F_0 \longrightarrow M \otimes_R N \longrightarrow 0$$

is exact. Hence the homology in positive degrees is zero, and therefore $\text{Tor}_1^R(M, N) = 0$ for every R -module N .

Conversely, suppose that $\text{Tor}_1^R(M, N) = 0$ for every R -module N . To show that M is flat, it suffices to show that $M \otimes_R -$ preserves injections. Let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ be a short exact sequence. The long exact Tor sequence contains

$$\text{Tor}_1^R(M, C) \longrightarrow M \otimes_R A \longrightarrow M \otimes_R B \longrightarrow M \otimes_R C \longrightarrow 0.$$

By hypothesis, $\text{Tor}_1^R(M, C) = 0$. Thus the map $M \otimes_R A \rightarrow M \otimes_R B$ is injective. Since tensoring is already right-exact, $M \otimes_R -$ is exact. Hence M is flat. $\square$

We now immediately see an easier proof for the free implies flat part of Proposition 1. Let F be a free R -module. Since Tor is independent of the choice of free resolution, we may choose the simplest free resolution for F :

$$\cdots \longrightarrow 0 \longrightarrow 0 \longrightarrow F \longrightarrow 0.$$

After tensoring with any R -module N , we get

$$\cdots \longrightarrow 0 \longrightarrow 0 \longrightarrow F \otimes_R N \longrightarrow 0.$$

We see that

$$\text{Tor}_1^R(F, N) = \frac{\ker(0 \rightarrow F \otimes_R N)}{\text{im}(0 \rightarrow 0)} = 0$$

for every N . By Proposition 2, F is flat.

Proposition 2 gives a clean characterization of flatness through homological algebra, but it is not very practical, since it requires checking $\text{Tor}_1^R(M, N)$ for every single R -module N . This is in practice impossible for most R -modules M , except the nicest ones, for which there is frankly not much to say. Hence, we want to refine the criterion that we have developed into something more concrete. In particular, we can make the key observation that an obstruction to flatness can be detected from short exact sequences of the form

$$0 \longrightarrow I \longrightarrow R \longrightarrow R/I \longrightarrow 0$$

where $I \subset R$ is an ideal.

Proposition 3. *Let R be a ring, and M an R -module. If $I \subset R$ is an ideal, then the multiplication map $I \otimes_R M \rightarrow M$ given by $r \otimes m \mapsto rm$ is an injection iff*

$$\text{Tor}_1^R(R/I, M) = 0.$$

The module M is flat iff this condition is satisfied for every finitely generated ideal.

Proof. We shall leave the proof as a guided exercise. See Problem 1 of the worksheet. $\square$

This refined condition now allows us to prove some nice facts regarding the interplay between flat modules and torsion.

Corollary 1. *Let M be a flat R -module. If $a \in R$ is a nonzerodivisor in R , then a is a nonzerodivisor on M .*

Proof. Let $a \in R$ be a nonzerodivisor in R and define $I = \langle a \rangle$. Notice that we have a map $\varphi : R \rightarrow I$ given by $r \mapsto ra$. This is obviously surjective, and it is also injective since $ra = 0$ implies $r = 0$. Hence, $R \cong I$ as R -modules. If M is a flat module, then Proposition 3 says that the multiplication map $I \otimes_R M \rightarrow M$ is injective, but under the identification

$$I \otimes_R M \cong R \otimes_R M \cong M,$$

the multiplication map becomes $m \mapsto am$, and the injectivity of this map is equivalent to saying that a is a nonzerodivisor on M . $\square$

Corollary 2. *If R is a PID, then M is a flat R -module iff M is torsion free.*

Proof. The forward direction is immediate from Corollary 1. Conversely, if M is torsion free, for every element nonzero $r \in R$, we have $rm = 0 \implies m = 0$ in M . Since R is a PID, every ideal of R has the form $\langle a \rangle$ for some $a \in R$. If $\langle a \rangle = 0$, then the condition for flatness in Proposition 3 is trivial since $0 \otimes_R M \cong 0$. If $\langle a \rangle$ is nonzero, then $a \neq 0$, and since M is torsion-free, the multiplication map $M \rightarrow M$ given by $m \mapsto am$ is injective since a is a nonzerodivisor on M . But this is exactly the multiplication map

$$\langle a \rangle \otimes_R M \rightarrow M,$$

after identifying $\langle a \rangle \cong R$ (as seen in the proof of Corollary 1). Hence, Proposition 3 implies that M is flat. $\square$

In the local Noetherian case, we can actually reduce checking the vanishing of Tor for every finitely generated ideal to checking only the unique maximal ideal of R , up to some small additional assumptions on our module M . This is made precise by the following:

Theorem 1 (Flatness for Local Noetherian Rings). *Suppose that $(R, \mathfrak{m})$ is a local Noetherian ring, and let $(S, \mathfrak{n})$ be a local Noetherian R -algebra such that $\mathfrak{m}S \subset \mathfrak{n}$. If M is a finitely-generated S -module, then M is flat as an R -module iff*

$$\text{Tor}_1^R(R/\mathfrak{m}, M) = 0.$$

Proof Sketch: The forward direction follows immediately from Proposition 2. Conversely, one shows by induction that $\text{Tor}_1^R(N, M) = 0$ for every finite-length R -module N . Then, using the Krull intersection theorem and the Artin-Rees lemma, one reduces the injectivity of $I \otimes_R M \rightarrow M$ for an arbitrary ideal I to this finite-length case. Hence M is flat by Proposition 3.

Most often, we use the following two corollaries in practice.

Corollary 3. *Let $(R, \mathfrak{m})$ be a local Noetherian ring and let $(S, \mathfrak{n})$ be a local Noetherian R -algebra with $\mathfrak{m}S \subset \mathfrak{n}$. Then S is flat over R if and only if*

$$\mathrm{Tor}_1^R(R/\mathfrak{m}, S) = 0.$$

Proof. This is just the case where $M = S$ in Theorem 1. □

Corollary 4. *Let $(R, \mathfrak{m})$ be a local Noetherian ring, and let M be a finitely generated R -module. Then*

$$M \text{ is free} \iff M \text{ is flat} \iff \mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0.$$

Proof. We already know that free implies flat and by Theorem 1 with $R = S$, the flatness of M is equivalent to $\mathrm{Tor}_1^R(R/\mathfrak{m}, M) = 0$. Hence, it only remains to prove that when M is flat and finitely generated over the local Noetherian ring R , M is free. We leave this proof as an exercise for the worksheet. □

Example 3. Consider $R = \mathbb{K}[x]_{\langle x \rangle}$ with the unique maximal ideal $\mathfrak{m} = \langle x \rangle$, and consider

$$M = R/\langle x \rangle \cong \mathbb{K}.$$

Then M is not flat over R by the free resolution

$$0 \longrightarrow R \xrightarrow{\cdot x} R \longrightarrow R/\langle x \rangle \longrightarrow 0$$

and Corollary 4. We leave the details to the exercises.

References

- [Eis95] David Eisenbud. *Commutative Algebra: With a View Toward Algebraic Geometry*. Vol. 150. Graduate Texts in Mathematics. New York: Springer-Verlag, 1995. ISBN: 978-0-387-94269-8.